

ECO101 – Lecture 6 Study Notes

Topic: Short Run vs. Long Run Total, Marginal, and Average Costs

These notes follow the lecture slide's terminology and cover the main concepts, important lines, and formulas.

1. Lecture Overview

- Relationship between a firm's output and costs in the **short run**.
- Derivation and interpretation of the firm's **short-run cost curves**.
- Relationship between output and costs in the **long run**.
- Derivation and interpretation of the **long-run average cost (LRAC) curve**.

2. Short-Run Cost

In the short run, some factors of production are fixed. To produce more output, the firm generally needs to employ more labour, increasing its costs. The lecture focuses on three cost concepts: **Total Cost, Marginal Cost, and Average Cost**.

3. Total Cost (TC)

Total cost is the cost of all the factors of production used by a firm.

- **Total Fixed Cost (TFC):** Cost of fixed factors. It does not change as output changes. Example: rent of knitting machines.
- **Total Variable Cost (TVC):** Cost of variable factors. It changes as output changes. Example: workers' wages.
- **Key formula:**

$$TC = TFC + TVC$$

Important graph idea: TFC is constant; TVC rises with output; therefore TC rises with output. The vertical distance between TC and TVC equals TFC.

4. Marginal Cost (MC)

Marginal cost is the increase in total cost resulting from a one-unit increase in output.

$$MC = \Delta TC / \Delta Q$$

Example from the lecture: if output rises from 10 to 13 sweaters and total cost rises from \$75 to \$100, then $\Delta Q = 3$ and $\Delta TC = \$25$.

$$MC = 25 / 3 = \$8.33$$

Think of MC as: “**How much extra cost does producing additional output create?**”

5. Average Cost

The lecture identifies three average costs:

- **Average Fixed Cost (AFC)** – total fixed cost per unit of output.
- **Average Variable Cost (AVC)** – total variable cost per unit of output.
- **Average Total Cost (ATC)** – total cost per unit of output.

$$AFC = TFC / Q$$

$$AVC = TVC / Q$$

$$ATC = TC / Q$$

Very important relationship: $ATC = AFC + AVC$

6. Shapes and Relationships of Cost Curves

- **AFC slopes downward:** the same fixed cost is spread over more units as output increases.
- **AVC and ATC are U-shaped.**
- **MC is U-shaped:** it initially decreases because marginal product increases, then eventually rises because of diminishing marginal product.
- The vertical distance between ATC and AVC equals AFC.
- **MC intersects both AVC and ATC at their minimum points.**

Key rule: if $MC < \text{average cost}$, the average cost falls. If $MC > \text{average cost}$, the average cost rises. This applies to both AVC and ATC.

7. Cost Curves and Product Curves

The lecture connects cost curves to product curves. At low output, increasing marginal product can lower marginal cost. Eventually, diminishing marginal product causes marginal cost to rise.

8. Shifts in the Cost Curves

The position of a firm's short-run cost curves depends mainly on **technology** and **prices of factors of production**.

Technology: A technological improvement increases marginal and average product of labour. The same inputs can produce more output, so production costs fall and cost curves shift downward.

Fixed-factor price increases: An increase in rent or another fixed cost shifts TFC, AFC, and TC upward, while AVC, TVC, and MC remain unchanged.

Variable-factor price increases: An increase in wages, gasoline, or another variable cost shifts TVC, AVC, and MC upward, while AFC and TFC remain unchanged.

9. Long-Run Cost

In the long run, a firm can vary both **labour and capital**. Therefore, **all costs are variable in the long run**.

Long-run cost depends on the firm's **production function**: the relationship between the maximum output attainable and the quantities of labour and capital used.

10. Production Function and Diminishing Returns

Production function: relationship between the quantities of inputs used and the quantity of output produced.

The lecture discusses diminishing returns:

- For a given plant size, the marginal product of labour diminishes as more labour is employed.
- For a given quantity of labour, the marginal product of capital diminishes as more capital is used.

$$MP_K = \Delta TP / \Delta K \text{ (holding labour constant)}$$

In words: marginal product of capital is the change in total product caused by a one-unit increase in capital, with labour held constant.

11. Short-Run ATC vs. Long-Run Cost

Each short-run ATC curve is U-shaped because, as labour increases, its marginal product initially increases and eventually diminishes.

In the long run, the firm can choose its plant size. The **economically efficient plant** for a given output is the one with the lowest average total cost.

12. Long-Run Average Cost (LRAC)

LRAC shows the lowest attainable average total cost for each output level when the firm can change both its plant and the quantity of labour.

Important ideas:

- A firm producing a given output at the least possible cost is operating on its LRAC curve.
- LRAC is made from the **cheapest portions of the short-run average cost (SRAC) curves**.
- LRAC is a **planning curve**: it tells the firm which plant and amount of labour to use at each output to minimize average cost.

13. Economies of Scale

Economies of scale occur when average total cost falls as output increases.

Therefore, when economies of scale exist, the **LRAC curve slopes downward**.

The lecture identifies greater specialization of labour and capital as the main source of economies of scale.

14. Diseconomies of Scale

Diseconomies of scale occur when average total cost rises as output increases.

Therefore, when diseconomies of scale exist, the **LRAC curve slopes upward**.

15. Constant Returns to Scale

Constant returns to scale occur when average total cost remains constant as output increases.

When constant returns to scale exist, the **LRAC curve is horizontal**.

16. Minimum Efficient Scale (MES)

Minimum efficient scale is the smallest output level at which long-run average cost reaches its lowest level.

In simple terms: **MES = the smallest output where the firm achieves its minimum LRAC.**

17. Exam-Friendly Formula Sheet

Concept	Formula / Key relationship
Total Cost	$TC = TFC + TVC$
Marginal Cost	$MC = \Delta TC / \Delta Q$
Average Fixed Cost	$AFC = TFC / Q$
Average Variable Cost	$AVC = TVC / Q$
Average Total Cost	$ATC = TC / Q$
Average-cost relationship	$ATC = AFC + AVC$
Marginal Product of Capital	$MPK = \Delta TP / \Delta K$ (labour constant)
Economies of scale	LRAC ↓ as Q ↑
Constant returns to scale	LRAC constant as Q ↑
Diseconomies of scale	LRAC ↑ as Q ↑

18. Quick Memory Map

- **Short run**: some inputs fixed → TFC exists.
- **Long run**: labour and capital can both change → all costs are variable.

- **MC:** extra cost of extra output.
- **Average costs:** cost per unit.
- **MC crosses AVC and ATC at their minimum points.**
- **Better technology** → lower production cost → cost curves shift downward.
- **Economies** → LRAC down; **Constant returns** → LRAC flat; **Diseconomies** → LRAC up.
- **MES:** smallest output at minimum LRAC.